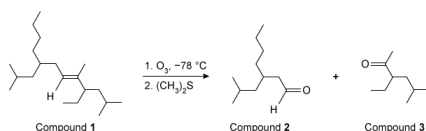


What are the IUPAC names of the ketone and aldehyde products, respectively, in the reaction below?

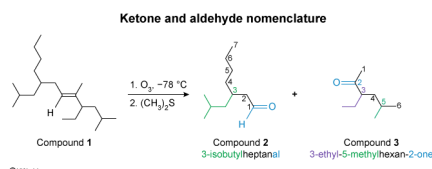


- ☐ A. 3-isobutylheptanal; 3-ethyl-5-methylhexan-2-one [39%]
- ☐ B. 3-butyl-5-methylhexanal; 4-ethyl-2-methylhexan-5-one [4%]
- ✓ ☒ C. 3-ethyl-5-methylhexan-2-one; 3-isobutylheptanal [52%]
- ☐ D. 4-ethyl-2-methylhexan-5-one; 3-butyl-5-methylhexanal [3%]

Correct

53% answered correctly

Explanation:



The name of an organic compound is determined based on the following steps:

1. Identify the **highest-priority** functional group, which determines the **suffix** of the compound name.
2. Determine the number of carbons in the **parent chain** (ie, the longest continuous carbon chain containing the highest-priority functional group). Number the chain to give the **lowest number possible** to the **highest-priority** functional group.
3. All other lower-priority functional groups are **named as substituents** and listed alphabetically at the beginning of the compound name. The number in front of each substituent name indicates its position in the parent chain (ie, the carbon atom to which the substituent is attached).

The highest-priority group in Compound **2** is an **aldehyde**, and the compound is named using the **suffix *-al***. The longest continuous chain containing the carbonyl (ie, the $\text{C}=\text{O}$ bond) is seven carbons (prefix *hept-*) and is numbered so the carbonyl carbon is carbon 1. The substituent on carbon 3 is an **isobutyl group** and is denoted in front of the parent chain name to give the name 3-isobutylheptanal.

The highest-priority group in Compound **3** is a **ketone**, and the compound is named using the **suffix *-one***. The longest continuous chain containing the carbonyl is six carbons (prefix *hex-*) and is numbered so that the carbonyl carbon has the lowest number possible (ie, carbon 2, in this case). The ethyl substituent on carbon 3 and the methyl substituent on carbon 5 are denoted in alphabetical order in front of the parent chain name to give the name 3-ethyl-5-methylhexan-2-one.

Therefore, the respective IUPAC names of the ketone and aldehyde products are 3-ethyl-5-methylhexan-2-one and 3-isobutylheptanal.

(Choice A) Although these are the correct names for Compound **2** (aldehyde) and Compound **3** (ketone), the names are given in the *reverse order* of what the question asks.

(Choices B and D) 4-ethyl-2-methylhexan-5-one **incorrectly numbers** the longest continuous carbon chain in the ketone to give the carbonyl carbon the highest number possible rather than the *lowest*. 3-butyl-5-methylhexanal incorrectly names the **parent chain** of the aldehyde.

Educational objective:

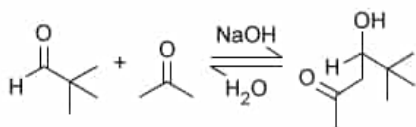
When naming organic molecules, the parent chain is numbered to give the lowest number possible to the highest-priority group. Ketones and aldehydes are named using the suffixes *-one* and *-al*, respectively.

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 403508

System:
Functional Groups and Their Reactions



What type of reaction is shown above?

- ☐ A. Transesterification [2%]
☐ B. Aldol condensation [35%]
☐ C. Fischer esterification [4%]
☒ D. Aldol addition [57%]

Correct

58% answered correctly

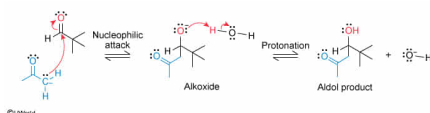
Explanation:

Acid- or **base-catalyzed aldol condensations** are important carbon-carbon bond-forming reactions in the synthesis of new molecules. The first step of an aldol condensation is an **aldol addition**, in which two carbonyl compounds (ketones and/or aldehydes) are joined together by a **nucleophilic addition** to form either a **β -hydroxy ketone** or aldehyde as the **aldol product**. If heat is applied to the reaction, **dehydration** of the aldol product occurs, forming an **α,β -unsaturated carbonyl compound** (ie, **enone** or **enal**).

For the given reaction, the ketone is the only reactant that has an α -proton (ie, an H atom next to the C=O bond). Consequently, the ketone is deprotonated by NaOH, forming a resonance-stabilized enolate.



A new carbon-carbon bond is formed when the enolate does a nucleophilic attack at the aldehyde carbonyl carbon, forming an alkoxide. Finally, the alkoxide is protonated to yield a β -hydroxy ketone as the aldol product.



Therefore, the reaction shown in this question is an **aldol addition** because the reaction yields an aldol product (ie, a β -hydroxy ketone).

(Choice A) **Transesterifications** occur between an **ester** and an alcohol, in which the alkoxy group on the ester is replaced by the alkoxy group on the alcohol.

(Choice B) The given reaction is *part of* an aldol condensation. An aldol condensation has a **dehydration step** after the aldol addition, which yields an α,β -unsaturated ketone or aldehyde.

(Choice C) A **Fischer esterification** occurs between a carboxylic acid and an alcohol, resulting in the formation of an ester.

Educational objective:

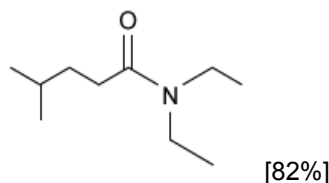
Aldol condensations are carbon-carbon bond-forming reactions between two carbonyl compounds (ketones and/or aldehydes). The first step (aldol addition) involves enolate formation and nucleophilic attack of a carbonyl, forming the aldol product, a β -hydroxy ketone (or aldehyde). The second step of an aldol condensation is dehydration of the β -hydroxy ketone (or aldehyde) to yield an α,β -unsaturated ketone (or aldehyde).

Time spent: 0 sec
 Subject:
 Organic Chemistry

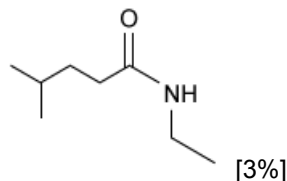
QID: 403509
 System:
 Functional Groups and Their Reactions

4-Methylpentanoic acid reacts with diethylamine using the coupling agent DCC to produce *N,N*-diethyl-4-methylpentanamide. What is the structure of the reaction product?

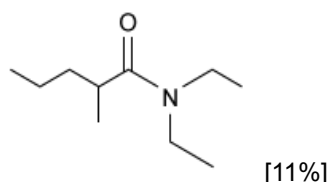
✓ ☒ A.



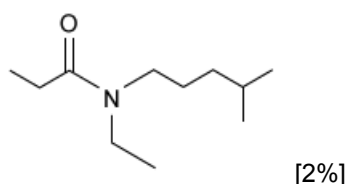
☐ B.



☐ C.



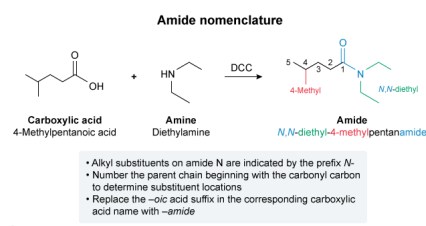
☐ D.



Correct

82% answered correctly

Explanation:



Organic molecules are named based on the **highest-priority** functional group in the molecule. The **parent chain** is identified as the longest continuous carbon chain containing the high-priority group, and it is numbered to give the lowest number possible to the high-priority group. The alkyl suffix from the parent chain is replaced with the high-priority **functional group suffix**. Lower-priority functional groups are **named as substituents** and listed alphabetically at the beginning of the compound's name.

Amides are **carboxylic acid derivatives** with a nitrogen bonded to the carbonyl carbon. Amides are named by replacing the *-oic acid* suffix in the corresponding carboxylic acid name with *-amide*. For **secondary and tertiary amides**, the **alkyl substituents** are listed before the parent chain name, with their **location indicated** by the **prefix *N*-**. If both alkyl substituents are the same on a tertiary amide, the *N* prefix is listed twice and *di-* is added before the substituent name. For example, an amide with two methyl substituents is named as an *N,N*-dimethyl amide.

The reaction described in the question forms the amide *N,N*-diethyl-4-methylpentanamide. The 4-methylpentanamide portion of the name indicates that the parent chain has five carbons (ie, prefix *pent-*) with a methyl substituent on carbon 4. The *N,N*-diethyl portion of the name indicates the amide N has *two* ethyl substituents. **Choice A** shows the only structure that matches the amide named in the question.

(Choice B) This structure has only *one* ethyl substituent rather than two.

(Choice C) The parent chain should be numbered beginning with the carbonyl carbon. The methyl substituent in this structure is on **carbon 2** rather than carbon 4.

(Choice D) The parent chain of an amide is the longest continuous carbon chain including the amide carbonyl carbon. This structure has the 4-methylpentyl portion as a substituent on the amide N rather than as part of the parent chain.

Educational objective:

Organic molecules are named based on the highest-priority functional group, and the parent chain is numbered to give the lowest number possible to the high-priority group. Amides are named by replacing the *–oic acid* suffix in the corresponding carboxylic acid name with *–amide*. The alkyl substituents on the amide N are indicated by the prefix *N–*.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403510

System:
Functional Groups and Their Reactions